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Indie Game Developer, UI/UX Designer, HCI Researcher

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Portfolios & Blogs: https://cidsho.github.io/

Location: Shatin, NT, Hong Kong

Language:
Cantonese (Native), Mandarin (Native), English (IELTS 8)

Self-Introduction

Interdisciplinary Experience Designer and Human-Computer Interaction Researcher, specializing in reconstructing experience design logic through data science. Grounded in profound research on Ludic Experience (game-inspired interaction), I adhere to a closed-loop design philosophy that integrates "behavioral data – cognitive models – interactive prototypes". Capable of extracting core experience demands from user/player behavior trajectories, emotional feedback, and system logs, then translating them into quantifiably validated design strategies. Possesses dual-layer design control spanning micro-interaction logic to macro-system architecture. Committed to reducing cognitive friction through cross-cultural insights (multilingual/multi-device/community culture adaptation), while ensuring commercial viability and emotional resonance in design solutions.

Education

Tongji University | 2021—2024
- Master of Arts in Design (Human-Computer Interaction)
- Weighted Average Score: 90.8/100
- Relevant Coursework: Interaction Design, Service Design, User Research

Zhongnan University of Economics and Law | 2017—2021
- Bachelor of Management
- Weighted Average Score: 83.06/100

Skills

- **Product, Prototype, Interaction Design:** Axure/Figma, Adobe Suite
- **Developing Skills:** Unity 3D (VR Interaction Design)
- **Others:** Digital Painting, Filming, Basic knowledge of HTML

Projects

- MTR XR Interactive Experience: Enhancing Passenger Interaction Through Contactless XR Technology — Feb 2024 — Apr 2025

  Roles: Research Assistant, Product Strategist, User Researcher

  Collaborators: Hong Kong Polytechnic University × MTR Hong Kong
  - **Tripartite Product Strategy (Government-Enterprise-University) Product Strategy:** Jointly with Hong Kong Productivity Council and technology vendors to complete requirement analysis. Led cross-departmental service framework design and project resource pre-allocation. Integrated technical feasibility studies with UX strategy, utilized AIGC tools for interactive prototype concepts and GUI wireframes. [Link]
  - **Solution Delivery:** Independently planned technical implementation pathways. Completed end-to-end user journey design (user journey maps, interface prototypes, dynamic previews), delivered solutions meeting dual criteria of feasibility and user value.
  - **Outcome Translation:** Proposal adopted by partners entering pilot preparation phase. Technical pathway analysis [Link] influenced subsequent technology procurement. UX design scheme established as benchmark for product iteration.
- ShadowPlayVR the Game/Creative Support Tool — 2022 — Present

  Roles: Interaction Designer, Unity3D Developer, UX Researcher
  - Design-Assist Workflow Application Reproducing Embodied Creative Experience via Motion Detection & Specialized Interaction Systems — For Professionals [Link] — 2022 — Present
    - **Situation:** Addressing difficulties in inheriting local intangible cultural heritage creation techniques from creators' perspective.
      - **User Value Needs:** Build applications compatible with existing workflows for users unfamiliar with digital tools/immersive environments (middle-aged/elderly artists), while providing rapid prototyping/preview functions for young artists.
      - **Social Value Needs:** Provide digital collaborative creation tools for intangible cultural heritage to promote content revitalization and sustainable industry development.
    - Tasks:
      - **Restore real-world experience:** Interaction methods align with physical creation habits (e.g., manipulation, real-time preview).
      - **Seamlessly integrate efficiency tools:** Introduce modern features (AI auto-modeling, rapid prototyping) while hiding technical complexity to avoid workflow disruption and maintain creative flow.
      - **Differentiated adaptation:** Provide low-barrier "learning-free" operation for elderly artists, offer advanced customization for young artists.
    - Actions:
      - Research-Driven Design:
        - **Data tracking & A/B testing:** Capture user decision points (e.g., sketch modification frequency, tool-switching paths), compare acceptance between traditional/digital workflows.
        - **Workflow segmentation:** Identify high-value digitizable phases (e.g., sketch-to-3D model conversion), focus on immersive media's differential advantages.
      - Technical Implementation:
        - **AI sketch modeling tool:** Based on YOLOv8 to recognize hand-drawn sketch features, auto-generate 3D model textures and bind IK relationships.
        - **Embodied interaction control system:** Map user motion coordinates to virtual model's IK skeletal system, achieving 1:1 "hand motion-shadow puppet control" mapping preserving traditional manipulation rituals. Added "AR environment projection" allowing artists to preview virtual effects directly on physical desktops, reducing cognitive load.
    - **Results: Deployed at Shanghai Qibao Shadow Puppetry Museum for 6 months, assisting 6 intangible cultural heritage artists. Increased artists' prototype iteration efficiency by 30%, rapid preview feature enabled exploration of new creative possibilities increasing prototype output by 40%. Findings published at major HCI conferences (see later), project later open-sourced.**
  - Game Design Promoting Creative Behavioral Expression via Specialized Interaction Systems — For General Users — 2022—至今
    - **Situation: 1.** VR applications/games face "homogenized interaction paradigms" **2.** Creativity-expression gameplay is underexplored and hard to evaluate
      - **Domain Design Limitations:** Traditional VR interactions over-rely on preset controls, limiting metaphorical/exploratory creative expression; existing systems prioritize functionality over quantifiable "creativity guidance".
      - **Design Challenge:** Break existing paradigms to build mechanisms that both inspire subconscious creative behaviors and systematically validate "creativity enhancement effects".
    - Tasks:
      - **Paradigm reconstruction:** Use "performance" as real-world gameplay basis to design new interaction language [Link], simulating real-world manipulation logic as behavioral commands in VR.
      - **Quantitative validation:** Establish player behavior measurement and entertainment evaluation system based on creativity-driven gameplay.
    - Actions:
      - **Metaphor deconstruction & interaction validity testing:** Conducted Wizard of Oz progressive prototype tests with 60+ subjects to filter core motion library. Validated interaction effectiveness via A/B testing + task-driven guidance across devices (Quest 3, HTC Vive Pro, Valve Index).
      - **Kinematic semantic compilation:** Built stylized virtual scenes/projection models in Unity, parameterized user behavior data tracking mapped to control inputs.
      - **Evaluation framework:** Created "creative flow state + creative output quality + user reflection/feedback" system to verify gameplay feasibility and behavioral guidance effectiveness.
    - **Results: Built VR-UI Design System with 107 spatial interaction controls [Link]. Demo exceeded 200 downloads in first week on Meta Store, formed 40-member Discord community. Published at major HCI conferences, cited as key case study in VR interaction/creativity-driven gameplay research.**
- Emotional Robotics Collaborative Project Technical Development | Product R&D Assistant — Tongji UXLab & Huawei Co. Coop. Project — 2022—2023
- Multimodal Audio Design in AR Environments | User Researcher — Tongji UXLab — 2022-2024

Peer-reviewed Papers/ Publications

- **Yihao He.** Not Just a Gimmick: A Preliminary Study on Designing Interactive Media Art to Empower Embedded Culture's Practitioner.
  Keywords: User Experience Research, Creative-support Application Design, Cross-platform Design Collaboration.
  Published in ACM SIGGRAPH Asia 2024
- **Yihao He.** ShadowRehearsal: Enhancing Traditional Art Design Workflows with XR Technology.
  Keywords: User Experience Research, Creative Workflow Service Design, Participatory Observation.
  Published in IEEE ISMAR 2024
- **Yihao He.** Architecting Social VR: Exploring Encounter and Serendipitous Interactions in Virtual Public Spaces.
  Keywords: Player Behavior Research based on Activity Hotspot Tracking, Virtual Space Design, Data-Driven Design Methodology.
  Published in HCII 2024
- **Yihao He.** Shadow Costar: Exploring Collaborative Performances in Virtual Reality.
  Keywords: Script-based AI, AI-Enhanced UX Design.
  Published in HAI 2023
- **Yihao He.** ShadowPlayVR: Understanding Traditional Shadow Puppetry Performance Techniques Through Non-Intuitive Embodied Interactions.
  Keywords: Embodied Interaction Design, Knowledge-Dissemination Application Design, Gaming Experience Structuring.
  Published in ACM VRST 2023

Awards

- Outstanding Student Scholarship (Grand Prize) | 2024
- Outstanding Student Scholarship (Grade 3) | 2023
- National College Student VR Competition (Excellent Award, Leader) | 2022
- Excellent Student Cadre | 2017-2019, 2021